Eric Hanchen Jiang

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RESEARCH INTERESTS

I build reliable, scalable LLM agents that reason, learn from experience, and operate over long horizons. My work spans (i) agentic systems for open-ended research; (ii) memory architectures that let agents accumulate and reuse knowledge across tasks and runs; (iii) efficient inference infrastructure for serving persistent agents at scale; and (iv) alignment and safety, with an emphasis on inference-time control and robust agent behavior.

EDUCATION

University of California, Los Angeles (UCLA) Aug. 2025 – Jun. 2029 (expected)
Ph.D. in Statistics and Data Science Advisors: Ying Nian Wu and Kai-Wei Chang

University of California, Los Angeles (UCLA) Jun. 2022 – Jun. 2025
B.S. in Statistics & Data Science; B.S. in Mathematics of Computation GPA: 3.85/4.00

RESEARCH EXPERIENCE

University of California, Los Angeles (UCLA) 2025 – Present
Graduate Researcher Mentors: Terence Tao, Amit Sahai, Kai-Wei Chang

- Co-developed **Continual Graph Memory (CGM)** [5], a persistent graph memory for mathematical agents to retain facts, state, and lessons across runs; CGM empowers mathematical research agents to solve multiple longstanding open problems, including Erdős Problem #561, Erdős Problem #289, Jamison Caterpillar Conjecture, and ect.
- Built **ansatze.ai**, an AI workspace for mathematical research that integrates agent runs, inspectable dependency graphs, Lean 4 verification, and human-in-the-loop feedback, making research workflows traceable and resumable.

Stealth Startup Jul. 2026 – Sep. 2026
Research Intern, LLM Inference Serving

- Developed **AgentKV** [7], a predictive KV-cache admission controller for long-horizon LLM agents that forecasts near-term context growth, maintains a lifecycle-safe working set under an adaptive HBM budget, and selectively persists KV cache to CPU memory.
- Achieved up to **3.7**× the throughput of stock vLLM on multi-GPU agentic workloads, and sustained roughly **75%** prefix reuse at 240 concurrent Qwen3.8-27B agents, versus at most 0.4% for stock vLLM.

Alibaba Cloud Group Jun. 2025 – Aug. 2025
Machine Learning Research Intern Mentor: Ranjie Duan

- Developed **Energy Landscape Steering (ELS)**, a fine-tuning-free, inference-time method that uses a lightweight external energy-based model to steer LLM hidden states away from false-refusal and jailbreak states, toward helpful or safely rejecting behavior.
- Raised compliance on the ORB-H over-refusal benchmark from **57.3% to 82.6%** while maintaining baseline safety across diverse models; published as a first-author paper at **ACL 2026** [3].

SELECTED PUBLICATIONS

First-Author Publications

[1] **Agent Q-Mix: Selecting the Right Action for LLM Multi-Agent Systems through Reinforcement Learning**
E.H. Jiang, L. Li, R. Sun, X. Liang, Y. Li, Y. Wu, H. Luo, H. Li, Z. Zhang, Z. Kang, et al. *COLM 2026*

[2] **Dynamic Generation of Multi-LLM Agents Communication Topologies with Graph Diffusion Models**
E.H. Jiang, G. Wan, S. Yin, M. Li, Y. Wu, X. Liang, X. Li, Y. Sun, W. Wang, K.-W. Chang, Y.N. Wu. *ACL 2026 (Main)*

[3] **Mitigating Over-Refusal in Aligned Large Language Models via Inference-Time Activation Energy**
E.H. Jiang, W. Ou, R. Liu, S. Pang, G. Wan, R. Duan, W. Dong, K.-W. Chang, X. Wang, Y.N. Wu, X. Li. *ACL 2026 (Main)*

[4] **SDSRA: A Skill-Driven Skill-Recombination Algorithm for Efficient Policy Learning**
E.H. Jiang, A. Lizarraga. *ICLR 2024 (Tiny Papers)*

First-Author Papers Under Review & Preprints *Equal contribution

[5] **Continual Graph Memory for Mathematical Research Agents**
J. Zhang*, J. Yu*, **E.H. Jiang***, J. Lu, Z. Zhang, X. He, H. Chae, E. Ji, A.K. Taylor, V. Sahai, Y. Kou, K.-W. Chang, R. Meka, N. Peng, A. Sahai, T. Tao, W. Wang. *Under review at ICLR 2027*

[6] **Memory as a Controlled Process: Learned Adaptive Memory Management for LLM Agents**
E.H. Jiang, Z. Zhang, Y. Wu, L. Li, D. Liu, X. Liang, R. Sun, Y. Li, E. Sun, H. Luo, Z. Kang, A. Caliskan, K.-W. Chang, Y.N. Wu. *Under review at ICLR 2027*

[7] **AgentKV: Predictive Admission Prevents KV-Cache Collapse in Agentic LLM Serving**
E.H. Jiang, J. Cai, Y. Lu, J. Pei, H. Zeng, J. Gesi, D. Wang, K.-W. Chang. *Under review at ICLR 2027*

[8] **From Solvers to Research: Large Language Model-Driven Formal Mathematics at the Research Frontier**
E.H. Jiang, X. Liang, Y. Zhang, Y. Wan, M. Li, H. Deng, A.K. Taylor, J. Baker, R. Raghavan, J. Zhang, et al. *Preprint, 2026*

Co-Authored Publications

[9] **Harness the Memory: A Holistic Evaluation of Memory Substrates in Memory Agents**
W.-C. Huang, W. Zhang, Y. Wu, Y. Chen, **E.H. Jiang**, W. Yang, Y. Yang, H.P. Zou, H. Zhang, Y.N. Wu, H. Wu, K.-W. Chang, P.S. Yu, X. Liu, A. Caliskan. *NeurIPS 2026*

[10] **Reasoning via Test-Time Instance-Level Policy Gradient in Latent Space**
H. Li, C. Li, T. Wu, X. Zhu, Y. Wang, Z. Yu, **E.H. Jiang**, S.-C. Zhu, Z. Jia, Y.N. Wu, Z. Zheng. *NeurIPS 2026*

[11] **Attention Sinks and Outliers in Attention Residuals**
H. Luo, H. Dai, S. Zhang, X. Chen, **E.H. Jiang**, Y. Li, J. Huang, C. Qiu, C. Xu, Z. Pan, H. Zhang, B. Wang, Y. Chen. *NeurIPS 2026*

[12] **RECAP: Looking Once Is Not Enough for Vision-Language Reasoning**
Z. Kang, T. Luo, C. Li, Z. Yu, F. Zhou, J. Qian, L. Wei, S. Chen, J. Li, Y. He, **E.H. Jiang**, R. Zhang, Z. Yao, H.L. Lee, G. Wang, et al. *NeurIPS 2026*

[13] **JPO: Juris Policy Optimization for Structured Legal Reasoning in Criminal Judgment Prediction**
Z. Kang, Y. Liu, T. Luo, L. Zheng, L. Wei, C. Zhu, J. Gong, J. Qian, **E.H. Jiang**, J. Liu, Y. Wang, H. Zhang, Z. Wang, R. Fu, Z. Lin, et al. *EMNLP 2026 (Main)*

[14] **When Seeing Overrides Knowing: Visual Dominance and Deferral-Based Method for Personalized Safety in VLMs**
E. Sun, Y. Wu, Z. Ma, **E.H. Jiang**, Y. Xiao, X. Yi, R. Krishna, et al. *COLM 2026*

[15] **Decoupled Alignment for Robust Plug-and-Play Adaptation**
H. Luo, J. Yu, W. Zhang, J. Li, C. Qiu, Y. Wang, **E.H. Jiang**, J. Hu, Y. Chen, et al. *COLM 2026*

[16] **Training LLMs for Divide-and-Conquer Reasoning Elevates Test-Time Scalability**
X. Liang, Z.Z. Li, Z. Lin, **E.H. Jiang**, H. Zhang, Y. Shen, K.-W. Chang, Y.N. Wu, et al. *ACL 2026 (Main)*

[17] **Beyond Magic Words: Sharpness-Aware Prompt Evolving for Robust Large Language Models with TARE**
G. Wan, L. Fu, H. Liu, Y. Jin, H.Y. Leong, **E.H. Jiang**, H. Geng, J. Bi, Y. Ma, et al. *ICLR 2026*

[18] **DAWN: Distributed LLM Multi-Agent Workflow Synthesis**
G. Wan, M. Zhou, Z. Wang, X. Shang, **E.H. Jiang**, G. Zhang, J. Bi, Y. Ma, et al. *AAAI 2026*

[19] **Multi-head Reward Aggregation Guided by Entropy**
X. Li, X. Chen, J. Fan, **E.H. Jiang**, M. Gao. *AAAI 2026*

[20] **CARES: Comprehensive Evaluation of Safety and Adversarial Robustness in Medical LLMs**
S. Chen, X. Li, M. Zhang, **E.H. Jiang**, Q. Zeng, C.-H. Yu. *NeurIPS 2025*

[21] **Statistical Guarantees for Lifelong Reinforcement Learning Using PAC-Bayesian Theory**
Z. Zhang, C. Chow, Y. Zhang, Y. Sun, H. Zhang, **E.H. Jiang**, H. Liu, F. Huang, Y. Cui, O.H.M. Padilla. *AISTATS 2025*

AWARDS & HONORS

UCLA Graduate Dean's Scholar Award (GDSA) 2025
Princeton University Graduate Student Fellowship (declined) 2025
Dean's Honors List, UCLA 2023–2024

PROFESSIONAL SERVICE

Conference Reviewer: NeurIPS 2025–2026, ICML 2026, ICLR 2026–2027, AAAI 2025–2027, ACL ARR 2026–2027
Journal Reviewer: Transactions on Machine Learning Research (TMLR), Stat, Electronic Journal of Statistics